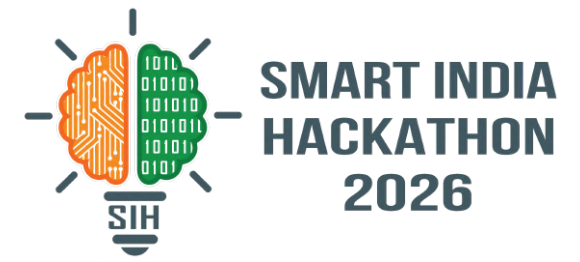
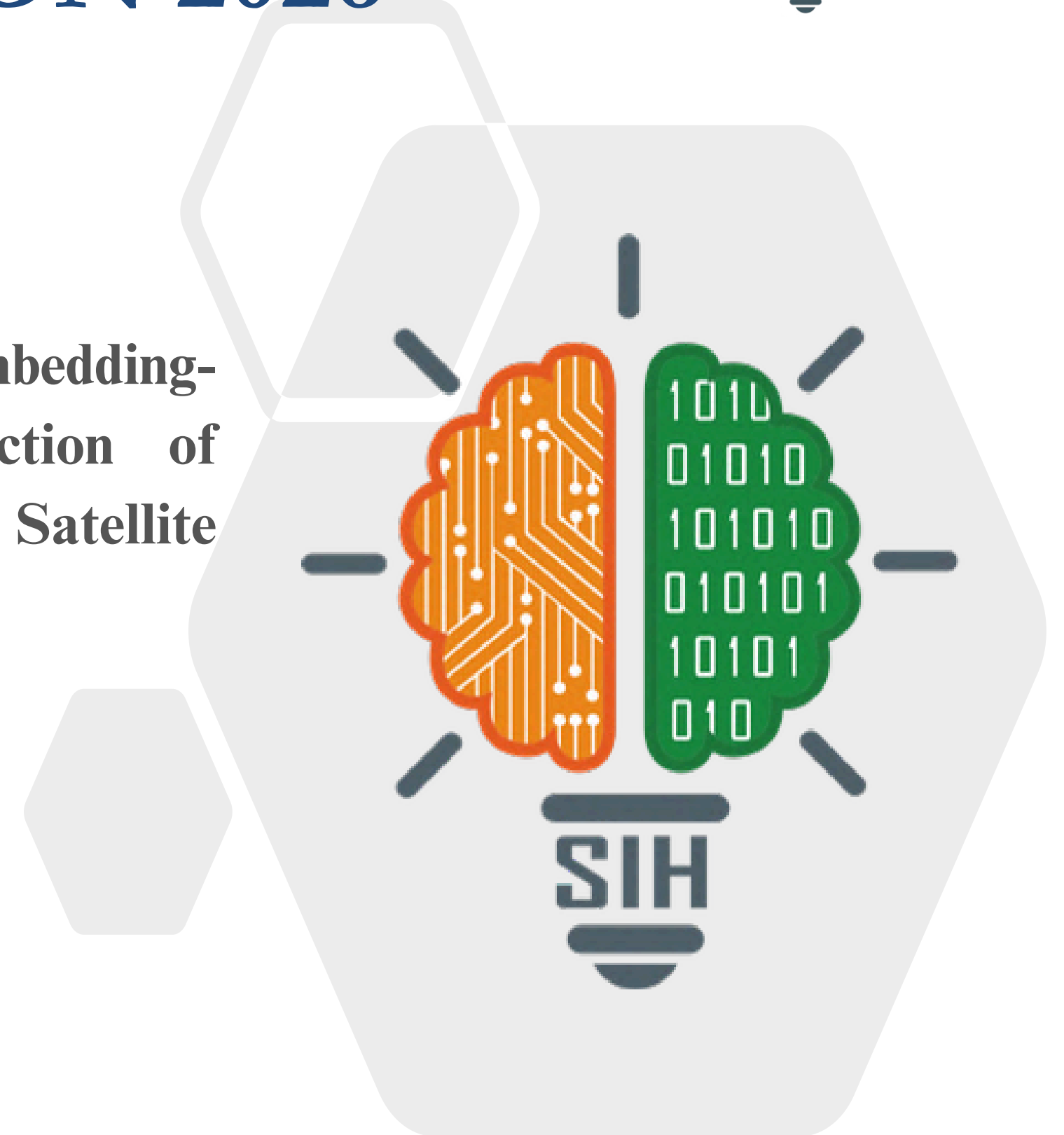


SMART INDIA HACKATHON 2026



- **Problem Statement ID – 26066**
- **Problem Statement Title- OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.**
- **Theme- Disaster Management**
- **Organization- Ministry of Earth Sciences (MoES)**
- **PS Category- Software**
- **Team ID- 141034**
- **Team Name - Bitminds**



! THE CRISIS



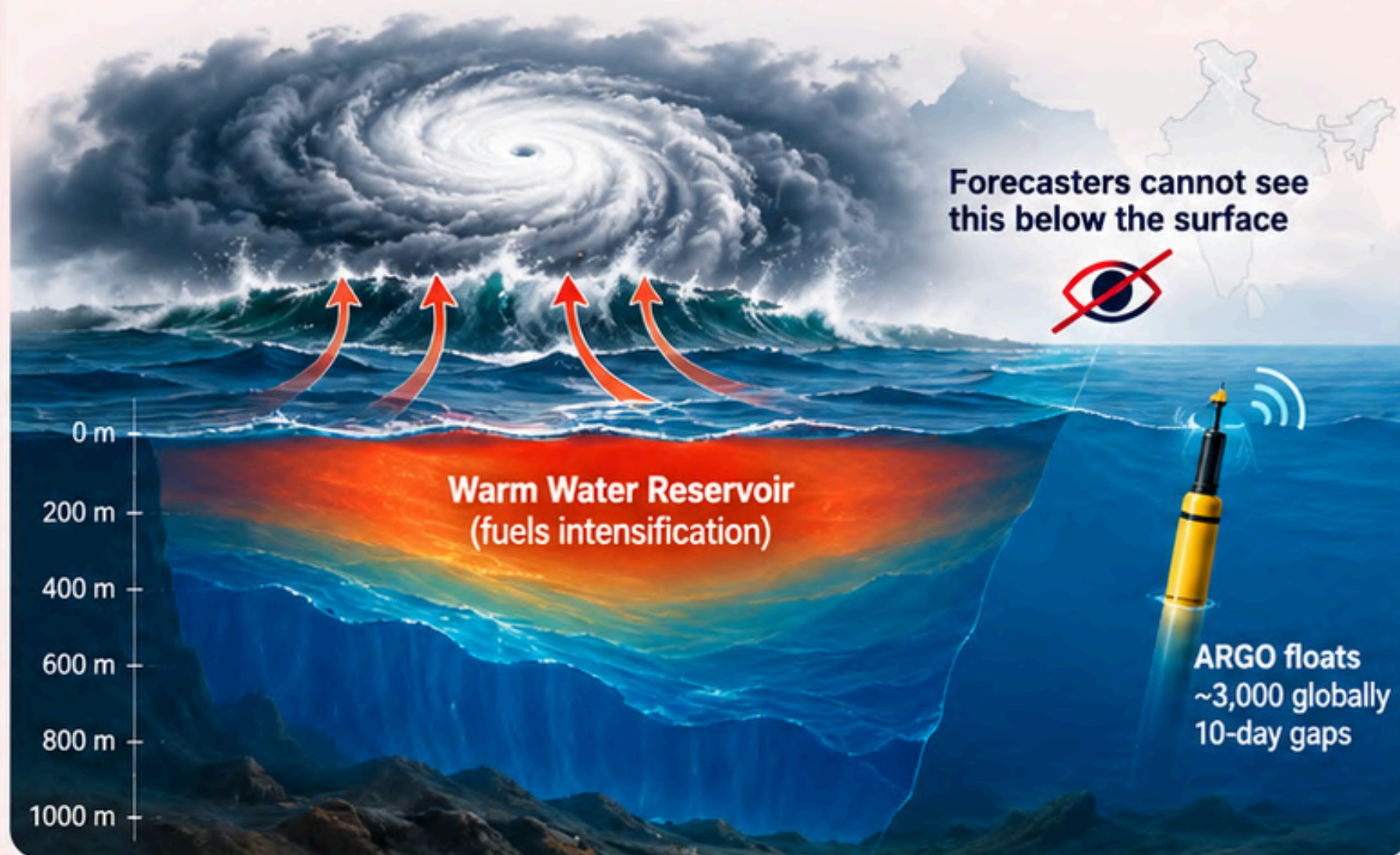
₹60,000 Cr+
Annual cyclone
damage



₹1.02 L Cr
Cyclone Amphan
damage



98 Lives
Lost in 2020



Forecasters cannot see
this below the surface

Warm Water Reservoir
(fuels intensification)

ARGO floats
~3,000 globally
10-day gaps



Sparse coverage
in North Indian Ocean
during cyclones



Blind forecasts
→ Late evacuations



Lost lives
and livelihoods



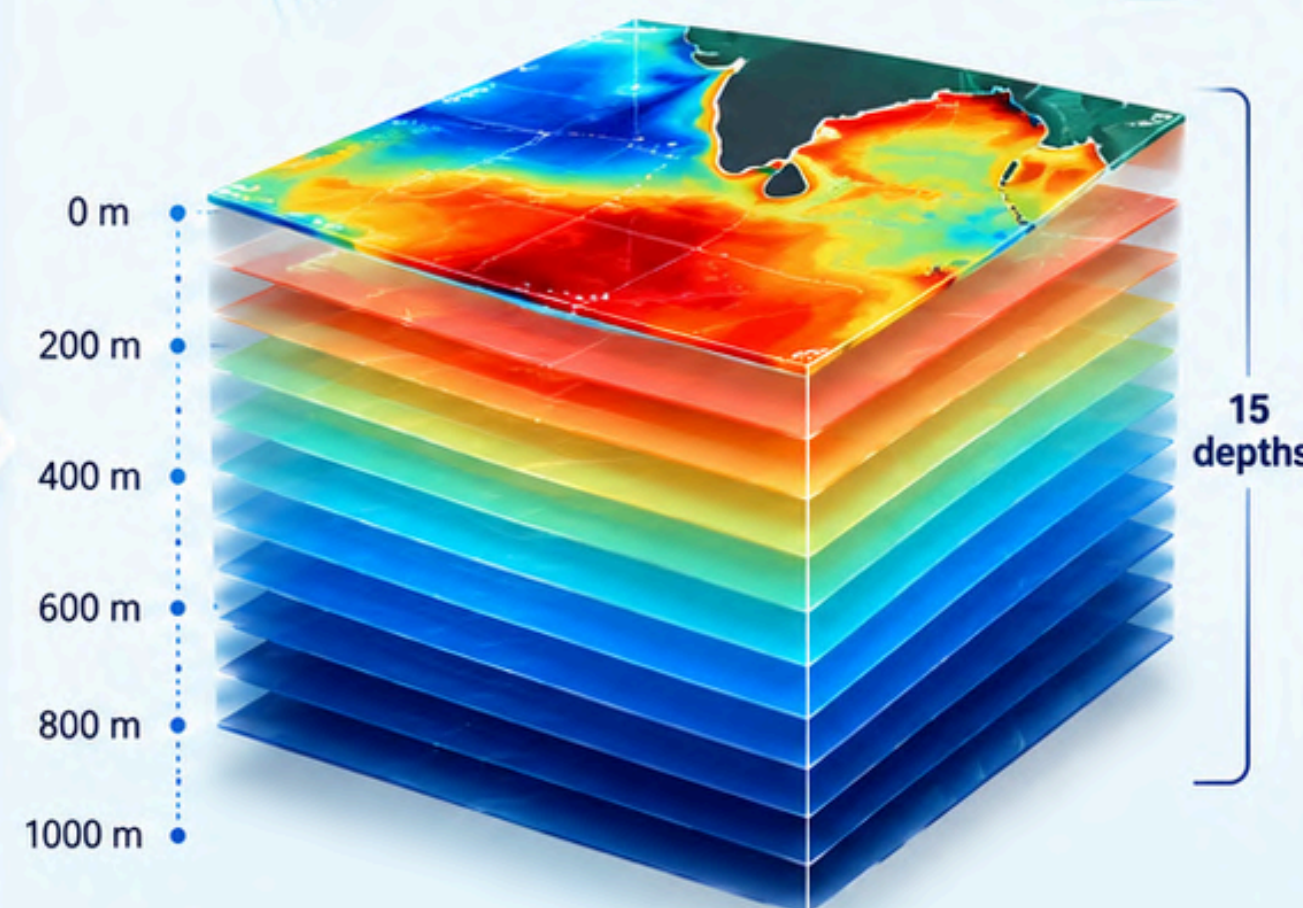
OCEANEMBED SOLUTION



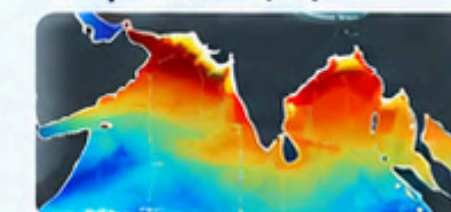
7 Surface Variables
SST, SSS, SLA,
Currents, Winds



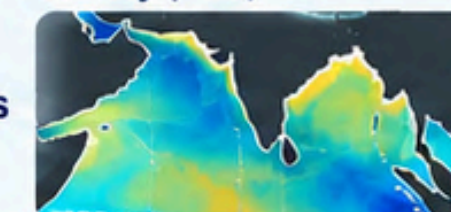
AI Reconstruction
3D Temperature + Salinity
+ Uncertainty



Temperature (°C)



Salinity (PSU)



Uncertainty ($\pm\sigma$)



Daily
Basin-Scale
3D Fields



200 ms
Neural
Inference



Physics-Informed
(Ekman, Coriolis,
TEOS-10)



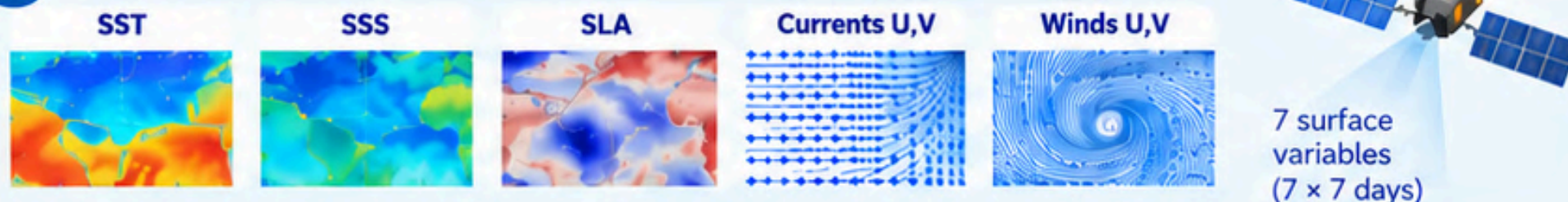
Real-Time
Cyclone Heat
Potential Maps



Calibrated
Uncertainty
($\pm\sigma$ per depth)

"From Satellite Skin to Ocean Depth — In 200 Milliseconds"

1 SATELLITE INPUT (7 × 7 days)



2 PHYSICAL PRE-PROCESSING



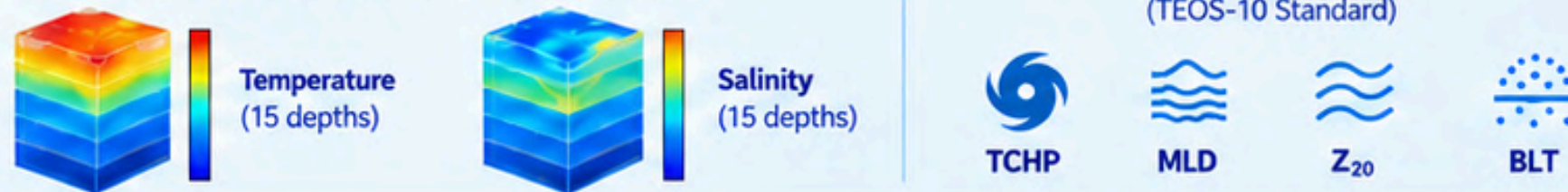
3 DUAL-STREAM ENCODER



4 DEPTH-ATTENTION DECODER



5 OUTPUT: 3D THERMOHALINE FIELDS



PERFORMANCE METRICS

**0.88°C**

Thermocline RMSE

Source: ARGO $\Delta t=0$ + GLORYS 2020–21 holdout**2.10°C**

Climatology RMSE

Source: 20-year GLORYS climatology baseline

**> 0.78**

CSS

Source: GLORYS 2020–21 holdout

**87.28%**

MSE Reduction

Source: Training convergence vs. initialization

**115.2 → 35.6**
kJ/cm²Cyclone Amphan TCHP
(79.6 drop)

Source: GLORYS holdout + RAMA validation

**0.107°C**ARGO Validation
(N = 1,420 profiles)Source: ARGO $\Delta t=0$, QC flags 1–2**142 ms**

Neural Inference

Source: NVIDIA T4 benchmark, batch=1

**38.4 s**

Full Pipeline

Source: End-to-end ingestion → export benchmark



TECH STACK

Frontend
Next.js 14Backend
FastAPI + Express.jsML
PyTorchData
Zarr, xarrayPhysics
gsw
(TEOS-10)Deployment
Docker



FEASIBILITY ANALYSIS

Practical. Scalable. Cost-Effective.

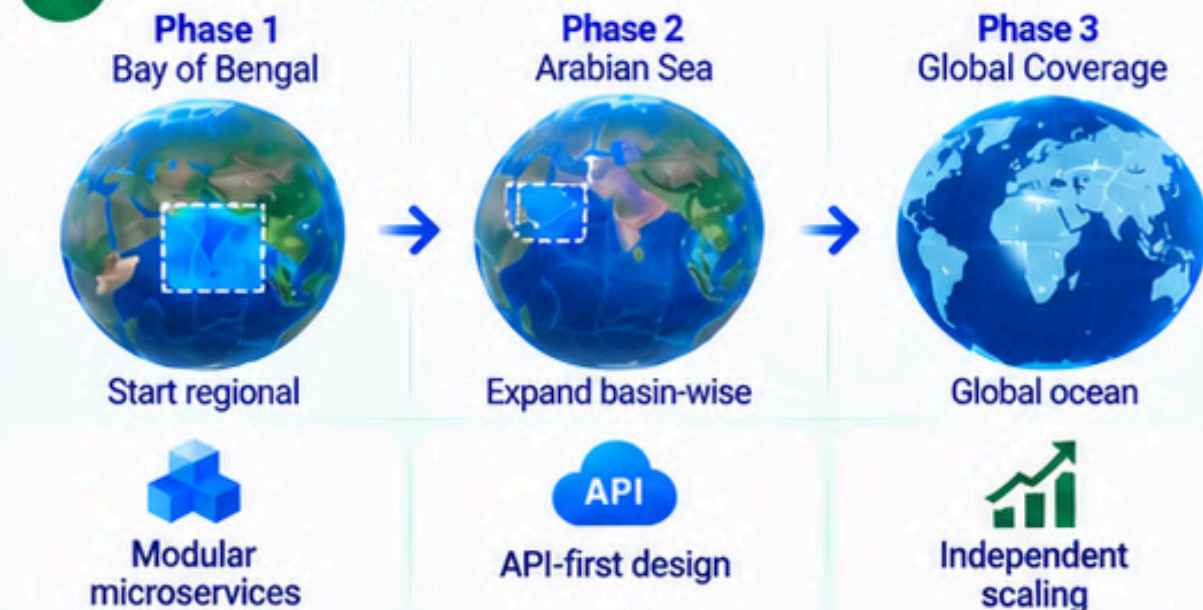
1 Infrastructure Reusability



- ✓ Existing Satellite Data
- ✓ Commodity GPU (NVIDIA T4)
- ✓ Integrates with INCOIS systems



2 Scalable Deployment



3 Cost-Effectiveness



VIABILITY PROOF

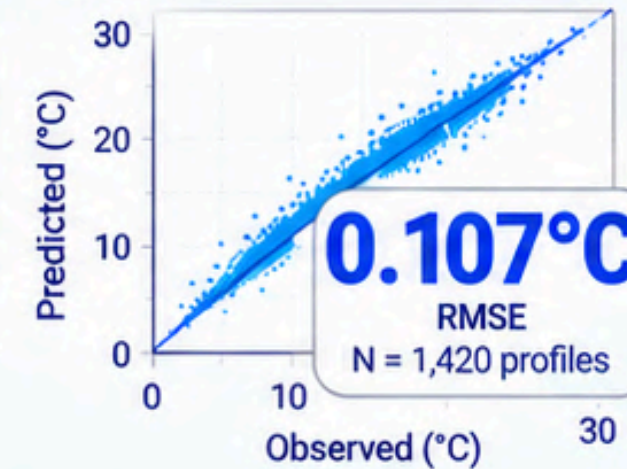
From Research to Real-World Impact.

1 Proven Concept

Validated against ARGO/RAMA



ARGO / RAMA Observations



2 Operational Readiness

Real-time, actionable insights



- 38.4 seconds**
Full pipeline
- Real-time dashboard
- LLM advisory copilot
- API for automated systems

3 Scientific Defensibility

Built on established ocean science



SUSTAINABILITY

High Impact. Low Footprint.



71.4%
lower energy
than numerical models



Single T4 GPU



Supports marine heatwave early warning



Long-term reusable for climate research



SECURITY & PRIVACY

Secure. Compliant. Transparent.



Secure Ocean Data
for a Safer Tomorrow



JWT authentication + rate limiting



Zod payload validation + sanitization



No PII collected (only geophysical data)



Open validation (no proprietary lock-in)



Compliant with INCOIS data sharing policies

"Trusted Data. A Safer Ocean."



MARKET CAP & ADDRESABLE OPPORTUNITY

Segment	Current	Target (2030)
 India Blue Economy	₹13.2 L Cr	₹24 – 30 L Cr (2035)
 India Ocean Tech	USD 13.2 Bn	USD 100 Bn
 Global Ocean Observation	USD 16 Bn	USD 30 Bn
 Maritime AI Market	USD 5.9 Bn	USD 64.7 Bn (2032)
 INCOIS Deep Ocean Mission	—	INR 4,077 Cr budget



40.8%
Maritime AI CAGR



>18%
Asia-Pacific Growth annually





KEY BENEFITS

MULTIPLE SECTORS, REAL IMPACT

-  **Cyclone Forecasting**
24-hour lead time (vs. 12 hours now)
-  **Marine Heatwaves**
Daily 3D thermal structure for early warning
-  **Fisheries**
Temperature and salinity fronts for productivity
-  **Shipping**
Route optimization with real-time ocean data
-  **Offshore Energy**
Safe operations with subsurface intelligence
-  **Climate Research**
Multi-decadal reconstruction for IPCC



Potential Annual Savings
₹10,000+ Crores



COMPETITIVE ANALYSIS

HOW OCEANEMBED COMPARES TO EXISTING APPROACHES

CAPABILITY	INCOIS EXISTING	ROMS / MOM6	STANDARD ML	OCEANEMBED
3D Temperature	Limited	Yes	Partial	✓ Full
3D Salinity	✗ No	Yes	✗ No	✓ Full
Daily Basin-Scale	✗ No	Yes	✗ No	✓ Yes
Real-Time Inference	✗ No	✗ No	Yes	✓ 142 ms
Physics-Informed	✗ No	Yes	✗ No	✓ Yes



Copernicus Marine Service <https://marine.copernicus.eu>



NOAA

NOAA

<https://www.ncei.noaa.gov>



NASA

NASA

<https://podaac.jpl.nasa.gov>



esa

ESA

<https://www.esa.int>



TEOS

TEOS-10

<https://www.teos-10.org>



EUMETSAT

EUMETSAT

<https://user.eumetsat.int>



Website Demo

[Click to view](#)



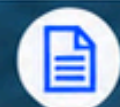
Youtube link : <https://youtu.be/GUdMyuLT980>



Github link : <https://github.com/Satyamkr7905/OCEANEMBED>



Website link : <https://oceanembed-phi.vercel.app/>



Report link : https://drive.google.com/file/d/1FT6h108DS1FdIU7_4tCzV7tmCN9Rhjk/view?usp=drivesdk

Dhruv Oberoi	:	Backend Developer & Leader	Dhanush Kumar NS	:	Data collection and cleaning
Satyam Kumar	:	Frontend Developer & Devops	Rajdeep Singh	:	Market Research and Analysis
Jashwanth P	:	Ml model Training	Sukhdeep Singh	:	PPT and Report